Odyssey

National cyber threat early warning

See the attack coming.

Public threat intelligence in. For every Indian organisation: which systems are at risk, how sure we are, what to do, and by when. On one laptop, with no cloud.

Final round

D3COY 1.0 · DJS ISACA · Problem Statement 01 · 1 Nov 2026

The warning was public. The decision was late.

16,730

new CVEs in October 2026 alone, from dozens of feeds, each needing a decision

72% → 21%

of new CVEs say which product and version they affect, January against now

29%

of the CVEs exploited in 2026 were attacked on or before the day they were announced

C-Edge, 2024

Jan 2024 Jenkins flaw disclosed; exploit public within days

31 Jul 2024 Ransomware through the supplier; about 300 banks lose UPI and ATM

19 Aug 2024 CISA lists the flaw, 19 days after the banks went down

Move the line left

Then: react after the breach



Flaw disclosed
public record, exploit code soon after

Nothing connects it
to who runs the product

Breach
the warning signs were public for months

Listed
the national list catches up weeks later

Now: act on the signal



12 days before
a scanning surge on the decoy grid (SIM) raises a pre-disclosure warning

Day 0
the CVE is matched to installed versions within seconds; a day-0 forecast flags the likely-exploited

Warning
tier, confidence, a countdown, a notice in the operator's language; a person approves

+20.5 days
median before CISA lists it, on real signals; up to 42

Every number recomputed live on the Proof page; leads from simulated sensors are reported apart from leads on real signals.

Every source, one record

Vulnerability databases

- NVD 2021 to 2026
- CISA KEV
- CISA SSVC
- FIRST EPSS

Advisories and context

- CISA ICS and alerts
- CERT-In style (SIM)
- MITRE ATT&CK
- NDMA SACHET

Telemetry, indicators, logs

- DShield honeypots
- Feodo C2
- OpenPhish
- Decoy grid (SIM)
- Firewall flows (SIM)

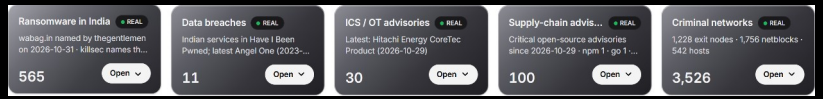
13 sources, 10 real, 3 simulated. Every item on screen says REAL or SIM. Also bundled, real: Exploit-DB, ransomware.live, Have I Been Pwned, GitHub advisories, Tor, Spamhaus DROP, Emerging Threats.

82,104 CVEs published in 2026

2,277 name a product an operator runs

354 Critical warnings, each to a named operator

In and out in the formats a SOC already runs: STIX 2.1, TAXII 2.1, CSV. A partner's indicators are matched against the record in seconds.



One advisory, start to finish

1

A real advisory arrives

CISA's ICS advisory for Siemens WinCC, read by the security LLM. It works even when the advisory carries no CVE id.

2

Versions parsed

Machine-readable ranges lifted from the text: vers:intdot/<8.0.0.11 and vers:all/*.

3

Exposure found

13 WinCC assets at 7 operators, checked against the inventory version by version. Patched hosts are marked safe.

4

Risk assigned

A score with its working, an SSVC tier, a CERT-In deadline. Patch where a fix exists, mitigation where none does.

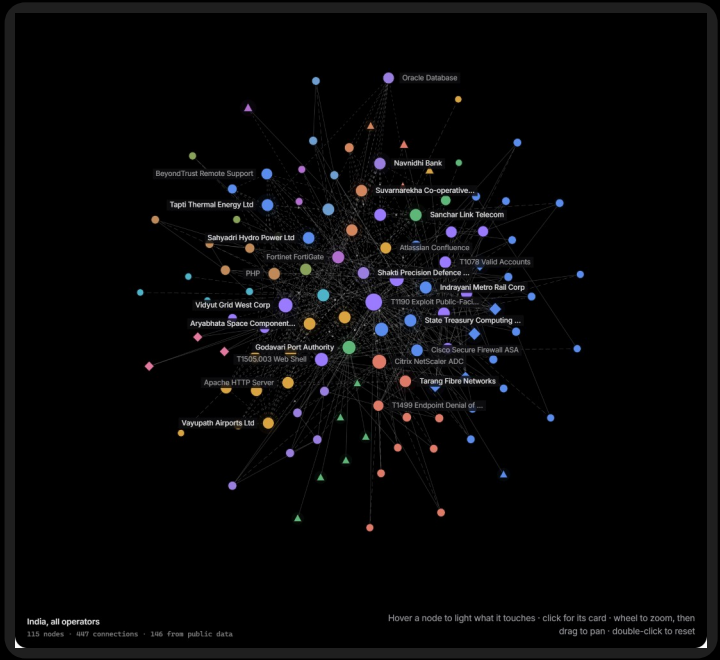
5

People decide

A notice proposed to each operator in its own language. An analyst approves. Every step lands in the audit trail.

The same path runs for every input. A KEV entry, a new CVE from NVD, a scanning surge on the decoys, a partner's indicator list, a disaster alert from SACHET: each is matched to who is exposed, scored, and turned into something a person can approve.

Patterns, not noise



Intel: every entity in the record and how they connect, 115 nodes, 447 links; fresh evidence pulses, old lies still

Reads A security LLM (MiST-8B) reads advisories and writes every brief. 22 of 23 briefs passed the fact validator; a sentence that fails is withheld, never shown.

Relates SecureBERT 2.0 links entities and related threats; a clustering model groups campaigns (12 active). Threat groups come from the ATT&CK record as overlap, never attribution.

Detects Scanning surges, C2 beacons and flow anomalies, every 5 seconds. A hostile advisory with hidden instructions is caught by the injection guard, 5 of 5, and still scored by the rules.

AI explains. Rules decide. No model sets a score, a tier or a deadline. All of it runs on one RTX 4070 laptop, 0 cloud calls; each job's model is set in a config file and was chosen by measurement on 2026 CVEs.

What goes first

Risk = 100 × Relevance × Importance × (55% Likelihood + 30% Severity + 15% Evidence)

Likelihood, from evidence, in rank order

On CISA KEV · SSVC says active · exploit seen on 2+ decoys	**1.0**
The vendor says exploited in the wild	**0.8**
A public proof of concept	**0.5**
A scanning surge on the decoy grid (decays with age)	**0.4**
EPSS ≥ 0.5 · CVSS ≥ 9 · automatable	**corroborate**
The day-0 forecast	**prediction**

Importance criticality tier 5 to 2; internet-facing ×1.2; blast radius up to ×1.3. **Relevance** the installed version checked against NVD, the vendor and the advisory; 0 false "affected" on the Ivanti case. **Confidence, scored apart** A to F source grades; agreement between independent sources raises it.

One warning, shown with its working

Tapti Thermal Energy · Ivanti Connect Secure 22.8R2.4 · CVE-2026-111357

Likelihood 0.55 × 1.0 — **45.8**
KEV 30 Oct · SSVC active · vendor: in the wild

Severity 0.30 × 1.0 — **25.0**
SSVC technical impact: total

Evidence 0.15 × 1.0 — **12.5**
CISA (A), NVD (A), vendor (B) agree

Relevance × 1.0
NVD and the vendor both list 22.8R2.4 as affected

Importance × 0.83
VPN gateway, internet-facing, tier 1

Risk **83.3**

Critical · SSVC: Act · 12 h to patch or mitigate (known-exploited, internet-facing) · chance of compromise 18%, no beacon seen

What to do, and by when

On every warning

1. **Patch** to a named fixed release, never a downgrade (22.8R2.5, 25.1.3.0 for the Ivanti case)
2. **Block** the indicators: STIX 2.1 export, risk lists for the firewall, C2 addresses refanged
3. **Investigate** a hunt playbook, ATT&CK mitigations, forensic triage where CISA asks for it
4. **Report** the CERT-In 6-hour report pre-drafted; RBI, SEBI, DPDP and NCIIPC clocks shown
5. **Notify** a notice in the operator's own language; a person approves before it goes

Odyssey proposes; a person approves anything consequential. Evidence-based notices, digests and hunts run on their own. Every action is in a hash-chained audit trail.

Nationally: the fewest changes that remove the most risk

93

changes remove half the national risk

205

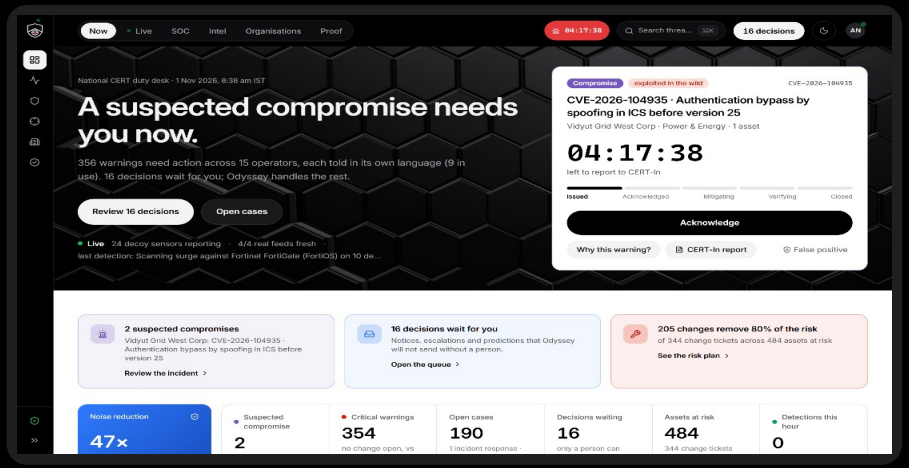
remove 80%, of 344 change tickets

One workflow per warning



Deadlines from the CERT-In Blueprint: 6 h for a suspected compromise, 12 h for a known-exploited flaw on the internet, then 24, 72, 120 h by tier. The countdown runs in real time on the warning.

One screen per question



Now: what needs me, with the AI brief and the score a click away



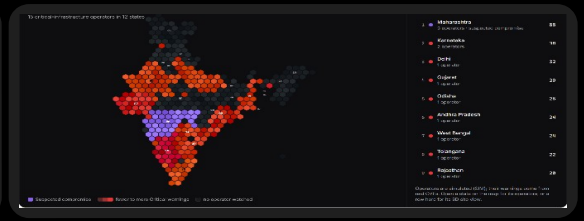
Intel: active threats, the technologies under attack, exposure by sector

Now · Live · SOC · Intel · Organisations · Proof

REAL or SIM on every panel. Feeds bundled, so nothing waits on the network. A night-shift mode. Every figure in a header recomputed live.



Where attacks start, and the operators they reach



India by Critical warnings; compromise in violet



A state in 3D: its agencies and the attack origins

It runs here. Offline.

0

internet or cloud AI calls, ever; the feeds are snapshot-backed

1

laptop: an RTX 4070 with 8 GB, every model under 8B parameters

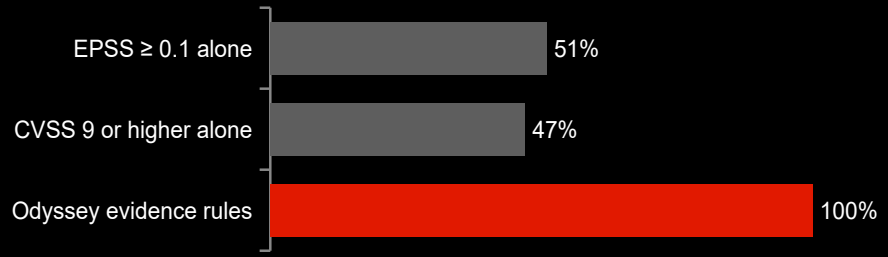
85 s

to re-score the nation: 15 operators, 612 systems, 2,277 flaws

221

engine tests green; 26 of 26 product flows; 25 of 26 browser steps

Share of 110 real exploited CVEs (CISA KEV, Aug to Oct 2026) each rule flags



A single score misses about half. KEV is one of our inputs, so 110 of 110 is by design; the honest test is the out-of-time forecast: 61.1% of later-exploited CVEs caught while flagging 2%.

Real, public: CVEs, KEV, SSVC, EPSS, CISA advisories, DShield, Feodo, OpenPhish, SACHET, Exploit-DB, ransomware.live, breach and network lists.

Simulated, labelled: the 15 operators and their inventory, the decoy grid, the firewall logs.

Proven on the wire: 45 of 45 real web attacks detected and blocked inline, 0 false positives, about 15 ms each. Models, scoring and workflow are explained on the Proof page; the test bench is a click away.

Real data. Measured claims. A human in the loop. Zero cloud calls.